Advanced Calculus I Non-hand-in Assignment 4

This homework comes from textbook and will **NOT** be collected (這份作業完成後不需要繳交、評分。). But for practice,or quizzes or exams, please make sure that you have idea to solve the following questions by yourself. (這些題目是很重要的練習，小考、期中、期末考會有類似的題目。練習這些題目的過程中所逐漸形成我們對於所學到的知識的"看法"，會在考試中很有幫助。所以，請一定要了解如何解題與解題背後的想法是什麼。)

Please prove the following propositions are true. (請證明下列命題爲眞。)

Section 9.1.
9.3 Remark (on page 303)
9.4 Remark and 9.5 Remark (on page 304)
9.7 Remark (on page 305)

第 4 週｜自主練習・課本題目補充

William R. Wade | An Introduction to Analysis | Fourth Edition, Global Edition

依教師 Non-hand-in Assignment 指定順序整理；以下為題目／命題敘述，不含解答。

Remark 9.3

Wade p.303

9.3 Remark. *For every* $\mathbf{x} \in B_r(\mathbf{a})$ *there is an* $\varepsilon > 0$ *such that* $B_\varepsilon(\mathbf{x}) \subseteq B_r(\mathbf{a})$.

Remark 9.4

Wade p.304

9.4 Remark. *If* $\mathbf{a} \in \mathbf{R}^n$, *then* $\mathbf{R}^n \setminus \{\mathbf{a}\}$ *is open and* $\{\mathbf{a}\}$ *is closed.*

Remark 9.5

Wade p.304

9.5 Remark. *For each* $n \in \mathbf{N}$, *the empty set* $\emptyset$ *and the whole space* $\mathbf{R}^n$ *are both open and closed.*

Remark 9.7

Wade p.305

9.7 Remark. *Statements ii) and iv) of Theorem 9.6 are false if arbitrary collections are used in place of finite collections.*

請針對無窮多個開集的交集、無窮多個閉集的聯集，分別構造反例。